

Homework 3

EE 999 — Introduction to Control Systems

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Due Wednesday, September 23, 2026

Problem 1 — Matlab

Code: ./hw_03_01.m

```
1 if exist('OCTAVE_VERSION', 'builtin')
2     pkg load control;
3 end
4
5 % Note these values were chosen to accentuate the differences
6 Kp = 1000;
7 Ki = 1000;
8 Kd = 1000;
9
10 %=====
11
12 m = 1000;
13 b = 50;
14 r = 1;
15
16 s = tf('s');
17 P_cruise = 1/(m*s + b);
18 t = 0:0.1:20;
19
20 %=====
21
22 C = pid(Kp);
23 T_P = feedback(C*P_cruise, 1);
24
25 %=====
26
27 C = pid(Kp, Ki);
28
29 T_PI = feedback(C*P_cruise, 1);
30
31 %=====
32
33 C = pid(Kp, Ki, Kd);
34
35 T_PID = feedback(C*P_cruise, 1);
36
37 %=====
38
39 figure
40 step(r*T_P, 'b', r*T_PI, 'r', r*T_PID, 'g', t)
41 axis([0 20 0 1.5*r])
42 legend('P', 'PI', 'PID', 'Location', 'southeast')
```

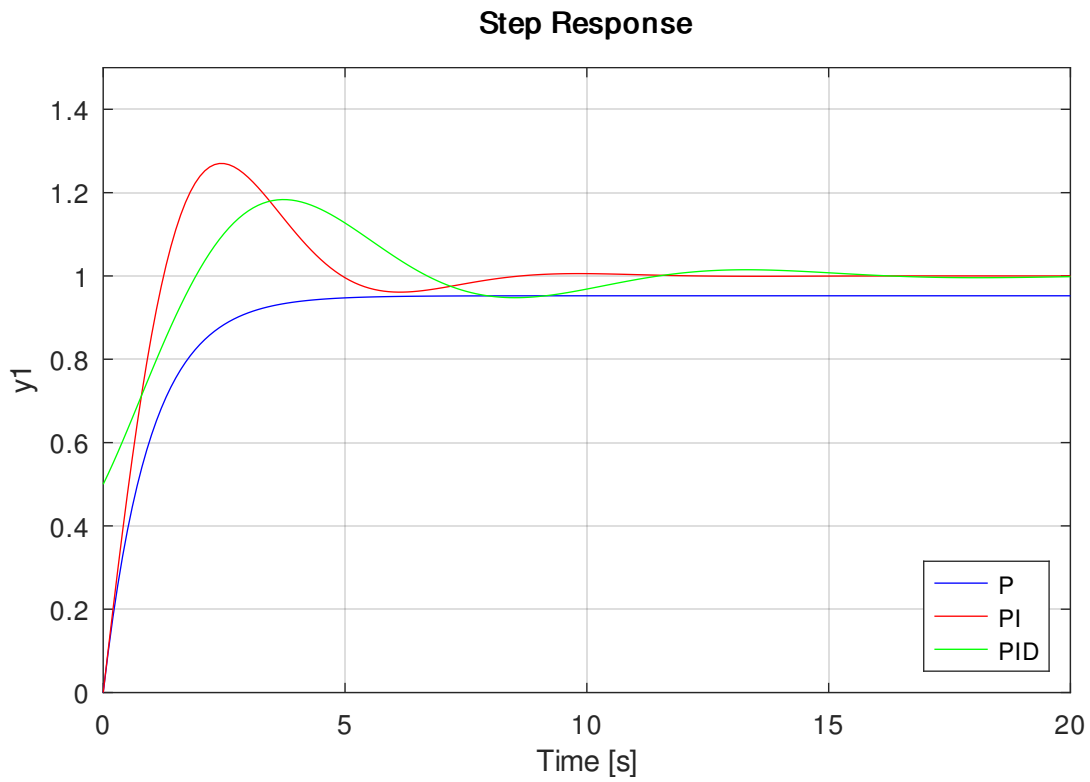


Figure 1: Step response of the cruise control system under P, PI, and PID control.

Problem 2.A — Realistic Op Amp

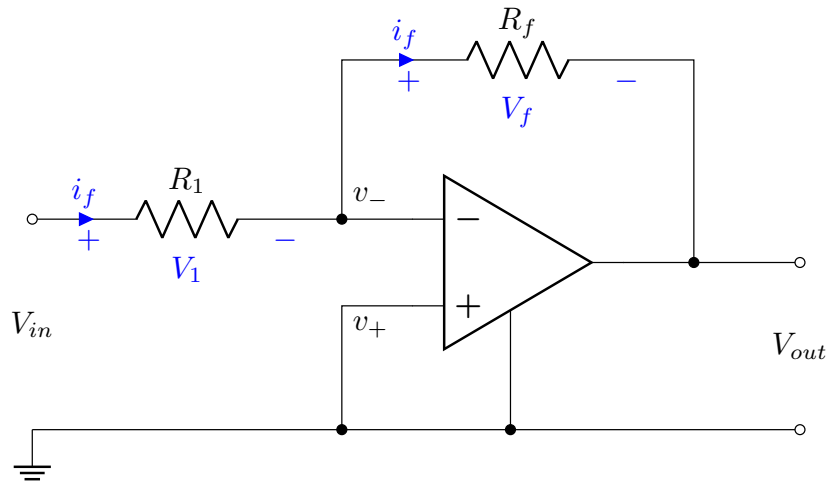


Figure 2: Inverting amplifier.

Start with the signals and parameters:

Symbol	Units	Description
v_{in}	[V]	Input voltage
v_{out}	[V]	Output voltage
v_1	[V]	Voltage across R_1
v_f	[V]	Voltage across R_f
i_f	[A]	Current through R_1 and R_f

Table 1: Symbols of Problem 2.A

Parameter	Units	Description
R_1	$[\Omega(\frac{V}{A})]$	Resistance
R_f	$[\Omega(\frac{V}{A})]$	Resistance

Table 2: Parameters of Problem 2.A

Symbols:

$$v_{in}(t) = v_{out}(t) + v_f(t) + v_1(t)$$

$$v_-(t) = v_{in} - v_1(t)$$

$$v_-(t) = v_{out}(t) + v_f(t)$$

$$v_1(t) = R_1 i_f(t)$$

$$v_f(t) = R_f i_f(t)$$

Introduce parameters:

$$v_{in} = v_{out} + R_f i_f + R_1 i_f$$

$$v_- = v_{in} - R_1 i_f$$

Transform:

$$V_{in}(s) = V_{out}(s) + (R_f + R_1)I_f(s)$$

$$V_-(s) = V_{in}(s) - R_1 I_f(s)$$

$$V_{out}(s) = -\frac{10^7}{s+1}V_-(s)$$

Substitute:

$$V_- = -10^{-7}(s+1)V_{out}$$

$$-10^{-7}(s+1)V_{out} = V_{in} - R_1 I_f$$

$$R_1 I_f = V_{in} + 10^{-7}(s+1)V_{out}$$

$$I_f = \frac{1}{R_1}(V_{in} + 10^{-7}(s+1)V_{out})$$

$$V_{in} = V_{out} + (R_f + R_1)\frac{1}{R_1}(V_{in} + 10^{-7}(s+1)V_{out})$$

$$V_{in} = V_{out} + \left(\frac{R_f}{R_1} + 1\right)(V_{in} + 10^{-7}(s+1)V_{out})$$

$$V_{in} - \left(\frac{R_f}{R_1} + 1\right)V_{in} = V_{out} + \left(\frac{R_f}{R_1} + 1\right)10^{-7}(s+1)V_{out}$$

$$R_1 V_{in} - (R_f + R_1)V_{in} = R_1 V_{out} + (R_f + R_1)10^{-7}(s+1)V_{out}$$

$$-R_f V_{in} = (R_1 + 10^{-7}(R_f + R_1)(s+1))V_{out}$$

$$G_{amp} = \frac{V_{out}}{V_{in}} = \frac{-R_f}{R_1 + 10^{-7}(R_f + R_1)(s+1)}$$

$$G_{amp} = \frac{V_{out}}{V_{in}} = \frac{-R_f}{10^{-7}(R_f + R_1)s + 10^{-7}(R_f + R_1) + R_1}$$